NPTEL MCQ Revision Notes — Lectures 31–40

Course: Sustainable Transportation Systems | **Instructor shown on slides:** Prof. Bhola Ram Gurjar, Dept. of Civil Engineering

Source: only the 10 uploaded PDFs (lec 31.pdf … Lec 40.pdf). No outside knowledge added. Tags like L33·s15 = Lecture 33, slide 15. Text in *[brackets]* is a note about the source. Yellow boxes = extraction issues. Text was extracted from all slides; picture/table/chart slides were rendered and read. Reference and "Thank You" slides carry no testable content.

Uploaded PPTs and contents

File	Lecture title	Slides
lec 31.pdf	L31: Sustainable Transportation Planning & Approaches-II: The Contemporary Planning Process (traffic calming)	64
lec 32.pdf	L32: Sustainable Transportation Planning & Approaches-III: Management Strategies (TSM)	50
lec 33.pdf	L33: Life Cycle Assessment (LCA): An Introduction	40
lec 34.pdf	L34: LCA: Theory & Practice (ISO 14040, types, Milan–Turin EV case)	34
lec 35.pdf	L35: LCA Case Study (Denmark biomass energy)	37
lec 36.pdf	L36: Material Flow Analysis (MFA)	32
Lec 37.pdf	L37: MFA Case Study (Vienna transport 2050)	33
Lec 38.pdf	L38: Concept of Circular Economy	41
Lec 39.pdf	L39: Circular Economy in Transport Sector (Kera, Finland)	36
Lec 40.pdf	L40: Modelling of Transport Emissions-I	57

Parts

Part 2 PPT-wise notes: L31, L32, L33, L34, L35, L36, L37, L38, L39, L40 (each starts on a new page, in this order)
Part 3 Rapid-revision sheet
Part 4 30 practice MCQs + answer key
Part 5 Coverage audit

L31 — Sustainable Transportation Planning & Approaches-II: Contemporary Planning Process

Planning background

- s3 **Common flaws in planning based only on forecasting & facility expansion** (graph labels): **A** early-stage forecasting (**under-estimation**); **B** late-stage forecasting (**over-estimation**); **C** "**Return to normal**" bias. (Graph also shows "forecasting reference year", "forecasting gap", "conventional growth cycle".)
- s4 **Main challenges** (UNESCAP 2015): non-inclusive development; difficult to meet growing demand (economic & population growth, rising income – faster than growth of economies); air pollution & GHG emissions growing faster than GDP; adverse health & social impacts; road safety.
- s5 **U.K. Transport Planning Society view** – planning should: prepare, assess, implement policies, plans, projects; improve & manage transport systems efficiently; plan at local, regional, national level; understand interrelationship between transport infrastructure and future shape of cities/towns; change people's perspectives/attitude and encourage alternative modes rather than private cars.
- s6 **Sustainable transportation:** capacity to support the mobility needs of a society in a manner that is the **least damageable to the environment** and **does not impair the mobility needs of future generations**.
- s7 Sustainable Development → Sustainable Transportation (linked with sustainable transport **Modes, Infrastructures, Operations**). Dimensions: **Environment** (climate change, air quality, noise, land use, waste); **Economy** (materials & energy, growth, employment, pricing, competitiveness); **Society** (safety, health, disturbance, access, equity).

The contemporary transport planning process

- s8 Commonly **scale-specific and multi-dimensional**. Traffic problems increased significantly **since the 1970s** despite much urban planning. **Early planning was based on induced demand** (to enhance capacities) rather than better management/maintenance of existing systems. **Preference: Accessibility rather than Mobility.**
- s9–10 **Multi-stage process (5 stages):** 1) **Goals & Objectives** – accessibility with improving safety & health, reducing vehicular emissions, enhancing economic opportunities, improving livability, promoting mobility by improving equity. 2) **Set of possible options** – what objectives are desirable & what they imply; several alternate scenarios considered. 3) **Identification of actors, institutions & stakeholders** – those affected, possible resolutions, **public participation**. 4) **Predicting outcomes, identifying benefits, assessing costs** – *critical step*; e.g. visual impacts, environmental dislocations, employment impacts. 5) **Choosing a course of action** – evaluating scenarios (costs & benefits); **political decisions**.
- s11 **Urban transport development paths:** development takes place within a **path dependency** → different levels of mobility & ownership (cars, buses, rail transit); once a path is elected, future development is **locked in**.
- s12 **Benefits of reducing car dependency:** healthier & active lifestyle; reduces demand for car parking; improves air & noise quality; improves safety on roads; reduction in traffic levels; provides improved alternative choices for travel; improves accessibility to destinations by all modes.
- s13–14 **Planning: dichotomy → continuum. Megalopolises** (mega-urban regions) = massive urban conurbations often encompassing several EMRs, structured along a corridor. Ex.: **BostWash** (Boston–Washington), **Tokaido** (Tokyo–Osaka), **Pearl River Delta** (Hong Kong–Guangzhou), Shanghai–Nanjing. Spatial imprint of transport varies with socio-economic & geographical characteristics; spatial interaction depends on circulation pattern of passengers & freight. Ex.: **North American cities** – urban form shaped by the automobile, dominance of highways.
- s15 City connectivity from the CBD (**CBD = Central Business District**); map legend groups: Production & Distribution (port, heavy industry, railyards, manufacturing, distribution clusters); Mobility & Accessibility (transit stations, shopping, airport districts); Transactions (financial/management districts).
- s16 Cities may be **Polycentric (more common)** or Monocentric. Traffic flows may be Organized or **disorganized (more common)**.
- s17 **Street networks:** Grid pattern was relatively **not common in 19th & early 20th century**. Grid: reduction in level of connectivity & level of landuse [as stated]. **Cul-de-sac** planning reduces/eliminates

through movements; cul-de-sacs = dead-end streets reducing extra traffic movement mainly in residential areas.

- s18 One-hour commuting by different modes: a **theoretical urban form** = summation of successive transport influences; assumes **uniform friction of distance and capacity** of transport infrastructure.

Summary of planning transitions (MIT notes) s19–20

#	From	To
1	Capital planning	Management & operations oriented
2	Long timeframes	Real-time control
3	Urban-scale planning & operations	Regional-scale planning & operations
4	Emphasis on mobility	Emphasis on accessibility
5	"One size fits for all"	Customer orientation, quality, pricing for services
6	Allocate capacity by queueing	Allocate capacity by pricing
7	Aggregate methods for demand prediction	Disaggregate methods for demand prediction
8	Public financing	Private and public/private partnerships in financing
9	Static organizations for institutional relationships	Dynamic organizations for institutional relationships
10	Emphasis on design of physical infrastructures	Emphasis on transportation as a complex large-scale integrated open system

Achieving sustainable urban transport planning s21–23

- Framework diagram (Gudmundsson 2014): **Governance** (political, institutional, regulatory) → **System** (infrastructure, energy, vehicles) and **Demand** (land use, economy, life style) → **Activity** (distance, mode, speed) → **Impacts** (economic, social, environmental).
- **Governance** of transport infrastructure is critical because of its strategic, economic & social importance. Interrelationship between **transport modes and infrastructures** should be considered in planning.

Transport Demand Management (TDM) s24–25

- Manage **both demand and the transport system**, not only demand as in earlier planning. High population densities favour walking, cycling, public transport. **Densification & integration:** planning concentrated along well-served transport corridors (**TOD – transit-oriented development**) and increasing densities in areas under rehabilitation.
- TDM is done through **(1) traffic calming measures** and **(2) traffic management strategies** (continued in L32).

Traffic calming measures s26–61

- **Main idea:** reduce speed of flowing traffic **or** reduce cut-through traffic by obstructing movements in one or more directions. Four general types: **Vertical deflections, Horizontal shifts, Narrowing roadways** (these three intended to **reduce speed**) and **Closures** (intended to **reduce cut-through traffic**).

Measure	Key facts from slides
Speed hump s27	Vertical obstruction; rounded & raised pavement; also "road humps/undulations"; slows cars to **about 30 km/h or lower** nearby; **not typically used** on major roads, bus routes, primary emergency response routes. (Fig. s29: hump = gradual raised area, low-speed roads, residential/local streets; 3–4 inches high, 10–12 feet; used on residential streets with speed limits up to 25 mph.)
Speed bump s28	Similar to hump but **more aggressive**; forces vehicles to slow much more than humps; used in **parking lots, residential streets** etc. (Fig. s29: abrupt raised area, private driveways/parking lots, private residential developments & shopping centers; 4–6 inches high, 1–3 feet.)
Speed table s30	Like humps but **long, raised, flat section in middle, ramps on ends**. At pedestrian crossing = **raised crosswalk**; only one direction of road = **offset speed table**.
Curb extension / bulb-out s31–33	Horizontal extension of sidewalk into street → narrower roadway; easier/safer pedestrian crossing (shortens curb-to-curb distance); at mid-block = **choker**. Gives space for benches & street lighting.

Diagonal diverter s34–36	Barriers placed **diagonally across four-legged intersections**, blocking through movements; also **full diverters / diagonal road closures**; can be partial closure allowing only **two-wheelers** and obstructing four-wheelers.
Lateral shift s37–38	Realignment of straight street so travel lanes shift in ≥1 direction. Suitable: local, collector, arterial roads; one-lane one-way & two-lane two-way; with/without bicycle facilities; **max appropriate speed limit typically 35 mph**; along bus transit routes.
Chicane s39–40	Series of **alternating curves/lane shifts** forcing motorist to steer back and forth (extra turns). Mid-block (entire block if short); speed limit **typically 35 mph or less**; one-lane one-way & two-lane two-way; open or closed (curb & gutter) cross-section.
Road closures s41–43	**Full closure:** barrier across street closing it to through traffic, usually leaving space for pedestrians/bicyclists; also called **cul-de-sacs, dead-ends, mini-parks**. **Half closure:** barrier blocks travel in one direction (creates one-way street) for a short distance on two-way street; also **partial closures / one-way closures**.
Median barrier / forced turn island s44–46	Raised islands along street centreline **continuing through an intersection**; blocks through movement from cross street; also called median diverter, intersection barrier, intersection diverter, island diverter. (Examples: with pedestrian refuge & bicycle cut-through.)
Median island s47–48	Raised island along centreline that **narrows travel lanes** at that location; same alternate names; separates opposing traffic and slows by briefly narrowing roadway; may have crosswalk.
Mini roundabout s49–50	Raised islands in **unsignalized** intersections around which traffic circulates; drivers slow to comfortably maneuver.
Roundabout s51–52	Raised islands **large in diameter**, unsignalized intersections; can serve high volumes of large trucks & buses depending on design; **different from traffic circles or mini-roundabouts**; possible substitute for traffic signal control; encourages steady flow; may have statues, fountains, community gardens.
On-street parking s53–55	Allocation of paved space to parking; narrows travel lanes, increases side friction; one or both sides; parallel or angled, but **parallel generally preferred** for maximized speed reduction.
Raised pedestrian crossing / intersection s56–57	Flat raised areas covering **entire intersections**, ramps on all approaches, often brick/textured; raised to sidewalk height → slows vehicles, easier crossing; also called **raised junctions, intersection humps, plateaus**; same-level crossing for all pedestrians & specially abled.
Road diet s58–61	Revision of lane use/widths → **one travel lane per direction** with minimum practical width, goal: reduce cross-section. Ex.: **four-lane two-way road → three-lane road** (two through lanes + centre two-way left-turn lane, **TWLTL**). May narrow lanes; alternate uses: bicycle facilities, left-turn lanes, on-street parking, raised medians, pedestrian refuge islands, sidewalks. Seattle Rainier Avenue: 4-mile dangerous lane, drivers at **40 mph+**; after modification speeds decreased by **approx. 95%** [as stated].

Conclusion s62

- Contemporary planning ensures sustainable development by emphasising **accessibility rather than mobility**; focus on **management of existing traffic & infrastructure** rather than mere capacity enhancement; traffic calming ensures safe planning by controlling traffic **speed and volume**; traffic management strategies should be combined with calming and policy measures.

Important Distinctions to Remember (L31)

- **Hump vs bump:** hump = gradual, residential/local, ~30 km/h; bump = abrupt, more aggressive, parking lots/driveways.
- **Speed table** = flat top + ramps; raised crosswalk (at pedestrian crossing) vs offset speed table (one direction only).
- **Curb extension** at mid-block = **choker**. **Full closure** (cul-de-sac/mini-park) vs **half closure** (one-way).
- **Lateral shift** (single realignment, ≤35 mph) vs **chicane** (series of alternating curves).
- **Mini roundabout** vs **roundabout** (large diameter; substitute for signals). Calming types: 3 reduce *speed*, closures reduce *cut-through traffic*.

L32 — Sustainable Transportation Planning & Approaches-III: Management Strategies

Transportation System Management (TSM) s3

- Solution to **peak-hour congestion, delay, accidents** in cities. Widening roads, elevated flyovers, bypasses, urban expressways **may prove costly**.
- **TSM:** a set of measures adopted to ensure the **most productive and cost-effective use of existing transportation services and modes**.
- Strategies covered (s2): prohibited turning; one-way streets; car pooling; staggering office hours; chartered buses; peripheral parking; internal CBD shuttle; transport rationing; congestion pricing; parking restrictions; day-time truck restriction; real-time travel information; contraflow lanes; exclusive bus lanes.

Strategies

Strategy	Key facts from slides
Turning restrictions s4–7	Can be established **only if the street is capable of alternate routing**. Prohibited right/left/U turns **increase saturation flow and capacity** at an intersection; **cannot solve congestion wholly**, but assists other techniques. Replaced by: **T-turn** = diversion of right-turning traffic to an **alternative intersection** capable of more traffic; **G-turn** = diversion of right-turning traffic **to the left before the junction**; **Q-turn** = diversion of right-turning traffic **beyond the junction**. Example: Menlo Park (Willows neighborhood, California) – table: total traffic on Durham Street (at Willow Rd) 300 (2017) → 78 (2019), **−74%**; right-turn Durham St→Willow Rd 218 → 17 (**−92%**, prohibited in 2019 data); Willow Road (at Durham) total 1,164 → 1,611 (**+38%**); right-turn Willow Rd→Durham St 5 → 4 (**−20%**).
One-way streets s8–9	Movement permitted in only one direction; improves flow, increases capacity, reduces delays; **least expensive strategy**; beneficial when combined with banned turns, signals etc. Example: Karol Bagh, New Delhi (one-way circulation plan).
Car pooling s10–13	Travellers share ride from/to common origin/destination. Types: **casual/informal**; **real-time (app-based)**; **van pooling: 7 to 15 passengers** sharing cost of a van as a daily mode. Reduces private vehicles, congestion, fuel cost. **Social benefits:** reduced vehicle miles travelled; reduced fuel consumption & GHG; reduced air-pollution impacts; cost savings for public agencies & employers. **U.S.:** car pooling is the **2nd most common travel mode to work after driving alone**; its mode share has decreased (chart: carpool 19.7% in 1980 → 9.0% at last plotted point; drive alone 64.4% → 76.3%) [read from graph]. Dale City example: left lanes open to all traffic; **right lanes only for buses and cars with minimum three riders**.
Staggering office & school hours s14–16	Employees choose office hours, **core time fixed** (ex. 10 A.M–7 P.M or 7 A.M–4 P.M); reduces peak-hour traffic & improves public transport conditions; same for school hours. Sample chart: flexible morning/evening hours keeping core time and **total weekly hours same**. **Linköping (Sweden)** cost-benefit of (a) one-hour-later school day vs (b) half-hour staggered start (million SEK/yr): later school day – benefit 8 / cost 8.5; staggered – benefit 8 / cost 11.3 (benefit items: buses not needed 8; costs: pupils, teachers, principal). Staggered scenario: 1/3 classes start half hour earlier, 1/3 half hour later, 1/3 same time as today. **1 SEK = 8.61 INR.**
Peripheral parking (park & ride) s17–18	Public parks private vehicle at **city outskirts** and uses public transport to city centre; integrate with car pooling, BRTS, metro; reduces fuel cost, avoids congestion tiredness. Example: Amsterdam.
Chartered (institutional) buses s19	Connect urban centres; trips from city centre to business parks, schools & colleges, government offices; reduce private vehicle inflow, congestion, fuel use; good connectivity to city centre needed.
Internal shuttle in CBD s20–21	**Uninterrupted** shuttle bus services on all routes within city; better connectivity; integration with park & ride increases demand; mobile apps & real-time information increase passenger interest. Example: National University of Singapore (NUS) institutional & shuttle buses.

Congestion pricing s22–25	**Demand-based**; encourages private vehicles to shift away from congested routes in peak hours; private vehicles using route in peak hours **fined heavily**. **Advantages:** reduces peak congestion; efficient use of system (alternate routes/public modes); smooth flow saves time & delay; **revenue generation**; fewer accidents; reduced emissions & idling. **Example 1 – London: cordon pricing**, charge for private vehicles entering Central London from **early 2003**; successful despite opposition. **Example 2 – Singapore:** limiting car purchases, high licence cost, electronic tolls on highways, cordon pricing in downtown (CBD pricing zone).
Parking restrictions s26–28	In city centre **indirectly reduce vehicle flow**. Benefits: increased automobile occupancy; decrease in person trips; faster travel & less delay; more public transport use; less air pollution; low ambient noise. **London = Controlled Parking Zone (CPZ):** all parking controlled by hours, only in designated bays; hours **Mon–Fri 7am–7pm; Saturday 7am–11am**. Chart: monthly CBD parking rates 2011 – highest **London ≈ $1,083.59**, lowest Boston ≈ $438 (others e.g. Zurich $822.15, Hong Kong $744.72, Tokyo $744.00) [read from chart].
Transport rationing s29–33	Restricts **number of vehicles or their movement**; examples: **no-drive day, driving restriction, odd-even schemes**. **Odd-even (Delhi):** rationing mechanism; in theory controls pollution by reducing traffic & congestion by restricting on-road vehicles per day; applied when ambient air quality deteriorated much. **Limitations:** followed in Paris, Rome, Bogota, Mexico City, Beijing; mostly failed after initial temporary success; people buy a cheap secondary car; low public transport capacity/poor infrastructure causes distress. **Only-pedestrian on weekends:** for very high footfall places; **Connaught Place** (more than a million footfall daily); NDMC planned to restrict vehicles Saturdays & Sundays in inner circle → trader protest → rolled back. **No-Car Sundays, Jakarta** (capital of Indonesia): two Sundays a month; "Car-Free Sunday"; concept gaining traction incl. India & Singapore.
Day-time truck restriction s34–35	Time restriction on heavy vehicles on certain roads in peak hours; separating heavy vehicles gives: less delay, enhanced safety, reduced congestion, noise & air pollution. **Delhi–Gurugram Expressway: 14 km** stretch, no truck entry **8–10 A.M and 5–10 P.M**.
Real-time travel information s36–38	Infrastructure: central data repositories fed by developers/travelers/media-3rd parties; data sources shown: traffic, events, parking, weather, signals, CAD, transit, infrastructure. **Benefits:** increase system efficiency & capacity; enhance mobility; improve safety; reduce energy consumption & environmental costs; increase economic productivity.
Contraflow lanes s39–43	Lane made **exclusively for peak-hour traffic in one direction**; better connectivity, less congestion/travel time. Ex.: single lane of two-way road towards business park is contraflow in morning peak, opposite direction in evening peak. Usually for **public transport, emergency vehicles, bicyclists**. Examples: contraflow bus lane; contraflow bicycle lane on one-way street; **Dublin** (allowing entry in opposite direction only for cyclists).
Exclusive bus lanes s44–46	Reserve a lane exclusively for buses; possible only where adequate carriageway width (**minimum 3 lanes**); **adjacent to curb** to board & embark; exclusive bus/cycle lanes may also be contraflow lanes. Example: **Ahmedabad**.

Taming the complexity / Conclusions s47–48

- Sustainable transportation possible only with **integrated effort** of individuals, families, organizations, governments at local, regional, state, national levels.
- TSM ensures productive & cost-effective solution to peak congestion, delay, accidents; right strategy or **combination** gives good results; reduces private vehicles, congestion, delays, fuel cost, emissions.

Important Distinctions to Remember (L32)

- **T-turn** (alternative intersection) vs **G-turn** (left before junction) vs **Q-turn** (beyond junction).
- **Contraflow lane** (peak-hour, opposite direction) vs **exclusive bus lane** (needs minimum 3 lanes, next to curb); both may be contraflow.
- **Car pooling** types: casual, real-time (app), van pooling (7–15). **Park & ride** = outskirts parking + public transport; **CBD shuttle** = uninterrupted shuttle within the city.
- **Congestion pricing** (fine for peak-hour use; London 2003 cordon; Singapore) vs **parking restrictions** (indirect reduction) vs **rationing** (odd-even; fails with second cheap car).

L33 — Life Cycle Assessment (LCA): An Introduction

Definitions & objectives

- s3 **LCA:** systematic analysis of environmental impact over the **entire life cycle** of a product, material, process or other measurable activity. LCA models the environmental implications of the many interacting systems that make up industrial production.
- s4 **Life cycle** of a product/service: overall span from **extraction of raw material** → application by user → **waste and recycle**.

Objective of LCA s5–8	Meaning	Slide example
Product improvement	Identify opportunities to improve environmental aspects of products/services at various life-cycle points	If transport is major pollution contributor, assess alternative fuel vs more polluting vehicle
Decision making	In industry, government, NGOs (strategic planning, priority setting, product/process design or redesign)	BS-IV → **BS-VI** standard vehicles in India to improve air quality
Environment planning	Selection of relevant **indicators** of environmental performance incl. measurement techniques	Which parameters assess human-health impacts, e.g. particulate matter, CO
Marketing	Environmental claim, eco-labeling scheme, environmental product declarations	Consumers prefer less-impact products; companies claim backed by LCA

System boundaries s9–19

- Defined by **cut-off criteria** (define parts/materials included in and excluded from the product system). Depends on study objective; often a combination of cut-off criteria; vary with life-cycle stages considered. **Four main options:** Cradle to Grave, Cradle to Gate, Gate to Grave, Gate to Gate.
- **Linear production model:** raw-material extraction → waste generation in linear fashion. **Circular production model:** includes **recovery of material after usage**; considered more environment-friendly than linear.

Boundary	Covers	Used to determine
Cradle to grave	Whole **linear** model: raw-material extraction up to waste generation	—
Cradle to cradle (**closed-loop LCA**)	Whole **circular** model; variation of cradle-to-grave, **exchanging waste stage with a recycling process** making it reusable for another/same product	—
Cradle to gate	Raw-material extraction through production phase (factory gate); assesses product only until it leaves the factory gates	Environmental impact of **production** of a product
Gate to grave	**Use and end-of-life** phases (everything post-production)	Impacts of product **once it leaves the factory**
Gate to gate	**Production phase only**	Impacts of a production step/process
Well-to-wheel	LCA of **transport fuels and vehicles**	More precise calculation/assignment of **GHG emissions & energy usage** for various fuels
EIOLCA (Economic Input-Output LCA)	Aggregates industry data to create impact data for specific sectors; **averages** sometimes used when no exact data	**Not precise enough** for decisions at product level

Well-to-wheel and EIOLCA are the "two other concepts used for special requirements" s17 .

Stages of LCA s20–21

1. **Defining goal & scope**
2. **LC inventory analysis (LCI)**
3. **Impact assessment**
4. **Interpretation**

LCA process diagram (s20, US EPA 1993): inputs = raw materials, energy → within system boundary: raw materials acquisition → manufacturing → use/reuse/maintenance → recycle/waste management → outputs = atmospheric emissions, waterborne wastes, solid wastes, coproducts, other releases.

1. Goal & scope s22–28

- **Scope items (8):** functional unit; reference flow; description of the system; system boundaries; allocation procedures; impact categories & impact assessment method; data requirements; data quality requirements.
- **Functional unit: quantified definition of the function** of a product. **Reference flow:** measure of product components & materials needed to fulfil the function defined by the functional unit; **all LCA data must be calculated/scaled per this reference flow**.
- **Data requirement:** energy inputs; raw material inputs; ancillary inputs; products & co-products; wastes; emissions to air, water and soil; other environmental aspects.
- **Factors affecting data quality (1/2):** data acquisition (measured/calculated/estimated? % primary vs secondary data from literature & databases); time-reference (when obtained, major changes since); geographical reference (country/region). **(2/2):** technology (state-of-the-art vs older); precision; completeness (missing data, how gaps filled); representativeness, consistency, reproducibility.
- **Allocation** = partitioning/relating inputs & outputs of a process to relevant products and byproducts. **By mass** (proportional to mass); **by heating value** (often used for production processes of **fuels**); **by market value**.

2. Life cycle inventory (LCI) s29–31

- Inputs & outputs of an industrial system are **measured and recorded according to the functional unit**; by end of phase an **inventory list** detailing all input/output data is created. Typically illustrated with a **flow model** (tables alone cause confusion).
- Example – **cement** (cradle-to-gate total per metric tonne of production): global warming potential (100 yr) **1040 kg CO_2-eq**; acidification **2.45 kg SO_2-eq**; eutrophication **1.22 kg N-eq**; tropospheric ozone formation **48.8 kg O_3-eq**; ozone depletion **2.61E-05 kg CFC11-eq**; non-renewable primary energy fossil **5250 MJ**, nuclear **345 MJ**; renewable (solar, wind, hydro, geothermal) **127 MJ**, biomass **165 MJ**.

3. Impact assessment (LCIA) s32–35

- **Definition:** evaluating the collected inventory data to make it meaningful in context of **potential damage to environment or human health**; impacts assessed by **impact category**.
- **Impact categories: Global warming** – rise in global mean temperature due to GHG emissions; **Acidification** – excess CO_2 → ocean pH decreasing, affecting marine life; **Eutrophication** – excess algae in water body impacting ecology; **Smog creation** – impacts human health; **Ozone depletion** – ozone layer depletes, UV filtration decreases → skin diseases.
- **US EPA LCIA steps (7):** selection & definition of impact categories → classification → characterization → normalization → grouping → weighting → evaluating & reporting LCIA results.
- Example table (s35): GWP – kg CO_2 eq; acidification – kg SO_2 eq; eutrophication – kg N eq; smog creation – kg O_3 eq; ozone depletion – kg CFC-11 eq. Outputs are converted into a few categories and impacts assessed.

4. Interpretation s36

- (i) **Identify significant issues** based on LCI and LCIA; (ii) **evaluate the study** – completeness, sensitivity, consistency, account for uncertainty; (iii) **conclusions, limitations, recommendations**.

s37 Overview figure: impact assessment (global warming, ozone depletion, summer smog, eutrophication, human toxicity, eco-toxicity, land use, resource consumption) ← life cycle inventory ← steps: raw material extraction, production of intermediates, production of main products, utilization, recycling/recovery/deposition.

Conclusions s38

- LCA = method to assess impact of product/service on environment; serves objectives from alternative selection to marketing; multi-step: goal & scope, inventory, assessment, interpretation.

> **Important Distinctions to Remember (L33)**

- **Cradle-to-gate** (production only up to factory gate) vs **gate-to-grave** (use + end-of-life) vs **gate-to-gate** (production phase only) vs **cradle-to-grave** (linear, whole) vs **cradle-to-cradle** (circular, closed-loop, recycling instead of waste).
- **Functional unit** (quantified function) vs **reference flow** (components/materials needed to fulfil it).
- **Allocation**: mass vs heating value (fuels) vs market value.
- **Well-to-wheel** = transport fuels & vehicles; **EIOLCA** = sector averages, not precise at product level.

L34 — LCA: Theory & Practice

ISO & ISO 14040 series s3–6

- **ISO:** independent, non-governmental international organization; membership of **165 national standards bodies**; develops voluntary, consensus-based, market-relevant International Standards; **officially came into existence in 1947**.
- **ISO 14040 series:** LCA procedure; quantitative assessment methods for environmental aspects of a product/service over its entire life-cycle stages. **ISO 14040 is the overarching standard encompassing all four phases of LCA**; various standards supplement it.

ISO standard	Description
14040	General principles, framework and requirements for LCA of products and services
14041	Guidance on determining **goal and scope** and conducting **life cycle inventory**
14042	Guidance on conducting the **LCIA** phase
14043	Guidance on **interpretation** of results
14047	Illustrative examples on how to carry out **LCIA**
14048	Information on **formatting of data** to support LCA
14049	Examples illustrating how to apply the guidance in **ISO 14041**

Types of LCA s7–12

Three types: **Conceptual, Simplified, Detailed** – each with strengths/weaknesses; application depends on context.

Type	Key facts	EV example (s12)
Conceptual	**Simplest**; basic assessment from a **limited, usually qualitative inventory**; shows which components/materials have largest impacts and why; only helps decision makers identify which products have a competitive advantage; also called **"Life cycle thinking"** (UNEP-DTIE, 2003)	Tail-pipe emissions for EVs are zero compared to substantial emissions by fossil-fuel vehicles
Simplified	**Screening** assessment covering the whole life cycle; uses **generic data and standard modules for energy production**; focuses on most important aspects/stages; **thorough assessment of reliability** of results. **Stages: Screening** (identify parts of life cycle/elementary flows that are important or have data gaps) → **Simplifying** (focus further work on important parts) → **Reliability** (check simplifying doesn't significantly reduce reliability)	General values of emissions from fuel-based vehicle manufacturing & operations compared with generic/secondary values of emissions from EV manufacturing, battery and electricity sources
Detailed	Full process; **extensive, in-depth data collection**; focuses on target of LCA; if only generic data available, data must be collected specifically on product/service. **Steps (Consoli et al. 1993):** planning, screening, data collection & data treatment, evaluation, improvement assessment	Exact data of all aspects of emission from mining of material up to disposal of battery or scrap calculated; assessment of benefits of EVs over regular vehicles

Case study: LCA of last-mile distribution logistics, Milan & Turin (Italy) s13–32

- **Study area: LTZ (Limited Traffic Zone)** of Milan and Turin urban centres (Italy, Europe). Turin = important business & cultural centre; Milan = important historical city, **second most populated in Italy after Rome**. Area: **Milan 8.2 km²; Turin 2.58 km²**.
- **LTZ:** regulatory intervention to improve air quality, lower noise, make area pedestrian-friendly; generally in high-footfall urban centres; vehicle traffic allowed only in limited off-peak hours or few vehicle categories exempted. LTZs = important area of analysis of delivery activities by logistics operators.
- **Details:** assesses environmental footprint of last-mile logistics by LCA; delivery activities in LTZ by a **large private logistics operator**; scenario analysis: current practice vs **replacing diesel fleet with electric vehicles (EV)**.

- **LCA steps followed:** functional unit & system boundaries → inventory of input/output items per life-cycle phase → assessment of potential impacts via specific indicators → interpretation of results (inventory analysis & impact estimation) relative to study objectives.
- **Goal – scenarios: Baseline** = typical day of a single large private operator, **exclusively diesel** vehicles; **ZEV** (Zero Emission Vehicle) scenario = diesel vehicles replaced by EVs. (EV = electric vehicles; LTZ = Limited Traffic Zone.)

Impact category	Unit	Description
Climate change	kg CO_2 eq.	Integrated heating effect in the next **100 years** from GHG emissions (CO_2, CH_4, N_2O etc.)
PM/smog	kg PM2.5 eq.	Emissions of particulate matter and its precursors (NOx, SOx, NH_3)
Photochemical ozone formation	kg NMVOC eq.	Volatile organic substances that, by action of light, promote oxidation reaction producing **ozone in the troposphere**
Acidification	Molc. H+ eq.	Lowering of pH of soils, lakes, forests due to acid substances (NOx, SOx) – "acid rain"

- **Data collected:** types of delivery methods (emission class, max load capacity); number of vehicles per type; kg of packages delivered per vehicle per day; km travelled per vehicle per day. **Observation period:** Milan **03/07/2017–30/09/2017**; Turin **21/04/2017–31/08/2017**.
- **Ecoinvent** = world's leading LCI database (transparency & consistency). Lorry classes (avg. load / GVW): **3.5–7.5 t: 0.98 / 4.98 t; 7.5–16 t: 3.29 / 9.29 t; >32 t: 15.96 / 29.996 t**. **Load factor** = ratio of average load to total vehicle freight capacity. GVW = gross vehicle weight.
- Diesel input data (s24): Lorry 3.5–7.5 t – max capacity 4.98 t, avg load 0.98 t, utilization 20%, fuel 0.1093 kg/tkm; Van type 1 – 1.25 t, 0.625 t, 50%, 0.2222; Van type 2 – 1 t, 0.5 t, 50%, 0.25. EV data (s25): Light – 730 kg, 24 kWh, 170 km; Heavy – 2000 kg, 28 kWh (1–3 batteries), 280 km (3 batteries).
- **Results (s26) change ZEV vs baseline:** Milan – climate change **−37%**, PM **+13%**, photochemical ozone **−47%**, acidification **−15%**; Turin – climate change **−49%**, PM **+4%**, photochemical ozone **−48%**, acidification **−24%**. [Slide 26 table shows −49% for Turin climate change; slide 30 text states 50% – both as in source.] (s27–28 bar/box charts: phases = delivery operations, fuel production, road maintenance, vehicle maintenance, electricity production, road construction, vehicle construction; ZEV with current mix, 25/50/100% solar.)
- **Why Milan ≠ Turin results (s29):** Turin fleet **exclusively EURO 5** whereas Milan combination of EURO 4 & EURO 5; diversity of routes in the two LTZs; different EV composition – Turin also includes **heavy ZEV vehicles** with lower specific consumption per kg of goods.
- **Conclusions (s30–31):** ZEV fleet cuts GHG by **37% in Milan, 50% in Turin**; with current electricity mix **particulate matter increases 13% in Milan, 4% in Turin**; NMVOC cut by almost **50%** in both. ZEV adoption should be combined with policies for a **decarbonized power system**. Sourcing **50%** of electricity from dedicated photovoltaic columns makes both scenarios' emissions comparable; **100% solar → particulate emissions −7%** (local vs regional emission perspective differs).

Conclusions s32

- LCA should be done according to **ISO standards**; various LCAs per study requirement; LCA efficiently used for **scenario comparison**.

Important Distinctions to Remember (L34)

- **14040** (overarching) vs **14041** (goal/scope + LCI) vs **14042** (LCIA) vs **14043** (interpretation) vs **14047** (LCIA examples) vs **14048** (data formatting) vs **14049** (examples for 14041).
- **Conceptual** (qualitative, "life cycle thinking") vs **Simplified** (screening; generic data; screening→simplifying→reliability) vs **Detailed** (extensive data).
- ZEV: GHG and NMVOC fall but **particulate matter rises** with current electricity mix.

L35 — LCA Case Study: LCA of Biomass-based Energy Systems, Denmark

Study area & background s3–6

- Denmark: Northern Europe; capital **Copenhagen** (cycle-friendly infrastructure); high-income economy.
- **Target: 100% renewable energy system in 2050**, primarily **wind power and biomass**, plus significant decrease in national energy demand. Realistic only through: significant reductions in energy demand; increased efficiency of fuel conversion technologies; higher share of wind power (e.g. **up to 50%**); replacement of fossil fuels with biomass; integration of transport sector into energy system (e.g. electric vehicles).
- **2008 sector-wise final energy consumption (pie): Transport 34% (highest)**, Households 29%, Industry 18%, Services/Commerce 13%, Agriculture 6%. [Slide text says households & industry follow with 29 & 18%.]
- **2008 energy sources:** fossil fuels ≈ **84%** of primary energy supply; only **16%** renewable (biomass, solar, waste); share of oil **39%**.

Scope, objectives, methodology s7–13

- **Scope:** environmental consequences of changing energy production; all **upstream, direct and downstream** emissions via life-cycle perspective; quantify impacts of future scenarios for **2030 and 2050**; compare with **reference year 2008**.
- **Objectives (3):** identification of potential biomass resources; selection of suitable biomass conversion technologies & efficiencies; quantification of environmental impacts of selected energy scenarios.
- **Goal:** assess environmental impacts of more sustainable energy scenarios; scenarios based on mix of residual agricultural resources, energy crops, wind & other renewables, and fossil fuels as needed; compare with 2008 reference.
- **Functional unit:** "**meeting the energy demand (electricity, heat and fuels) required in Denmark**" for 2008, 2030, 2050; does not describe amount of energy but ensures future living standards same as 2008 baseline.
- **Scope set-up:** LCA modelling for **Primary Energy Supply (PES)**; results normalized with PES for each year; impacts quantified with **time horizon of 100 years**.
- **Assessment approach:** select scenarios from national energy strategies & political targets → select conversion technologies & data → balance energy supply and demand per scenario → impact assessment & discussion.
- **Impact categories:** global warming (CO_2 eq); land occupation; acidification (area of unprotected ecosystem per unit energy); aquatic eutrophication with **two sub-categories – nitrogen & phosphorous**.

Energy scenarios s14–20

Five scenarios	Notes
Year 2008 (baseline); Year 2030	—
2050 – **CSV**	Conservative scenario (with fossil fuels)
2050 – **RME**	Biodiesel as **Rapeseed Methyl Ester**, totally produced from rapeseed
2050 – **BtL**	**Biomass-to-Liquid** and **Fischer–Tropsch** technology (except RME share equal to today's production in DK)

Last three (2050) represent alternatives for transport-fuel production. Abbreviations on diagrams: PJ = Peta Joule (**one PJ = 10 to power 15**, as stated), NG natural gas, CO coal, PV photovoltaic, E electricity, H heat, L losses, BP byproducts, F fuel, GH geothermal heating, ST solar heating, Syn syngas, BG biogas, PE petrol, DS diesel (2030 includes **4.5 PJ of RME biodiesel**), BE bioethanol, AF aviation fuel, BD biodiesel, SF synthetic fuel. Flow diagrams (s15–20) are Sankey-type energy-flow figures [numeric values not transcribed in full; e.g. 2050 CSV oil 76.0 → DS 30.3, BD 4.5, L/BP 17.3, AF 33.5; RME rapeseed 74.0, oil 35.0 → BD 34.8, L/BP 40.8, AF 33.5; BtL willow 76.2, rapeseed 9.6, oil 35.0 → BD 34.8, L/BP 52.6, AF 33.5].

Measures for energy demand s21–25

Area	2030	2050	Policy change / effect
Electricity production (SOFC = Solid Oxide Fuel Cell; CHP = Combined Heat & Power)	SOFC CHPs = 33% of total CHP plants; geothermal ≈ **4 TWh** (15% of demand in big cities)	SOFC CHPs = **100%** of CHP plants	Reduces primary energy supply from fossil fuel; decommissioning of old power plants; integration of geothermal production with incineration plants
Land transport	Passenger transport growth (**18%**) covered equally by train transport and "avoiding transport"; bicycling & walking meet **10%** of demand (5% in 2008); vehicles meet **60%**, railroads **30%**; bioethanol ≈ **5%** of passenger transport; **half of goods transport** diverted to ships & trains	Growth (vs 2030) "avoided"; vehicles **50%**, railroads **40%**; bicycling & walking **10%**; no bioethanol needed	Reduction of fuel consumption & fossil primary energy; **road pricing** and better physical urban planning; railroad expansion (higher capacity & speed); loading stations; EVs also needed as **storage system** for electricity fluctuations
Aviation	Domestic flights reduced to **5%** of current demand	Same as 2030	Railroad expansion; improved fuel-combustion technology of planes
Water transport	Ships lower fuel demand by **40%** vs 2008	By **60%** vs 2008	Improved fuel-combustion technology of ships

s26 **Biomass → energy technology (table):** manure & grass – anaerobic digestion (biogas & digestate; biogas to heat/electricity; digestate on land); wood & willow, straw – gasification (syngas & biochar); waste – incineration (electricity & heat); rapeseed – transesterification (RME, glycerin, solid biofuel); willow (BtL, FT) – gasification & **Fischer–Tropsch** (FT-biodiesel & biochar); straw (BE) – straw refinery (bioethanol, molasses, solid biofuel). s27 Transport energy totals (PJ): 2030 – E 24.1, BD 4.5, DS 72, PE 27.5, BE 4.7, AF 37.9; 2050 – E 43.3, BD 34.8, CH_4/SF 11.8, AF 33.5.

Impact results s28–31 (charts)

- **Global warming** (Gg CO_2-eq/PJ_PES): 2008 = **68**; 2030 = **38**; 2050 CSV = **20**; RME = **31**; BtL = **17**. (PP power plants; LUC = dLUC + iLUC; BtE biomass-to-energy.)
- **Acidification** (10^6 m^2/PJ_PES): 2008 = **301**; 2030 = **193**; CSV = **121**; RME = **177**; BtL = **135**.
- **Eutrophication-N** (Mg N/PJ_PES): 2008 = **14**; 2030 = **16**; CSV = **15**; RME = **29**; BtL = **24**.
- **Additional land occupation** (10^6 Δm^2/PJ_PES, vs 2008 base): 2030 = **624**; CSV = **861**; RME = **2089**; BtL = **1787**.

Study conclusions s32–35

- Significant GHG/global-warming reductions achievable **per PJ** of energy supplied.
- **Residual domestic biomass was insufficient** → extra cultivation of **energy crops** → significant environmental loads in most categories.
- **Large impacts of upstream land use changes (LUCs) made fossil diesel for heavy transport favorable to RME- and FT-biodiesel.**
- High aquatic eutrophication = direct consequence of energy-crop cultivation (increased fertilizer use). **Main means for GHG savings:** reduced energy demand, increased wind share, replacing fossil fuels with residual domestic biomass.
- **Main environmental challenge:** supply of biofuels for **heavy terrestrial transport, ships, defense and aviation**. Energy crops → impacts in global warming (mainly LUC), aquatic eutrophication, land occupation. Fossil diesel preferable to biomass fuels **except global warming where FT-biodiesel performed slightly better**.
- LCA gives clear objective understanding of impacts of choices; strong tool to compare policy alternatives; widely usable in transport policy & planning.

Important Distinctions to Remember (L35)

- **CSV** (conservative, fossil) vs **RME** (rapeseed biodiesel) vs **BtL** (biomass-to-liquid, Fischer–Tropsch).
- Fossil share 84% / renewable 16% / oil 39% (2008). Transport 34% > households 29% > industry 18%.

- Fossil diesel preferable to biofuels in most categories, **except global warming** (FT-biodiesel slightly better).

L36 — Material Flow Analysis (MFA)

Concept, definition, history s3–8

- MFA is based on the **law of mass conservation**: matter can change form through physical & chemical changes but is conserved; mass going into a system = **mass coming out + mass absorbed (stored) in the system**. Law dates from **Antoine Lavoisier's 1789** discovery (mass neither created nor destroyed in chemical reactions).
- **MFA definition** (Brunner & Rechberger, **2004**): systematic assessment of the **flows and stocks of materials within a system defined in space and time**.
- **Equation: Input = Output + Stock.** Example: 100 L fuel × 2 g lead/L = **Input 200 g**; **Output 150 g** lead in air; **Stock = 200 − 150 = 50 g** (remains within system).
- **History:** conservation of matter (input = output) first postulated by **Greek philosophers more than 2000 years ago**; mass-balance principle applied in medicine, chemistry, economics, engineering, life sciences. **Two original application areas: metabolism of cities**; **analysis of pollutant pathways** in regions such as watersheds or urban areas.
- **Objectives (1/2, 2/2):** delineate system of material flows & stocks; reduce system complexity while maintaining basis for decision-making; assess relevant flows & stocks quantitatively (check mass balance, sensitivities, uncertainties); present results in reproducible, understandable, transparent fashion; use results as basis for managing resources, environment, wastes; monitor accumulation/depletion of stocks, future environmental loadings; design environmentally-beneficial goods, processes, systems. Examples: hazardous substance causing health issues – create flow & stock of incoming/outgoing/stocking; PM10 vehicular emission – study environmental loading in a city region.

Key terms s9–13

Term	Definition (as on slides)
Material	Substances and goods
Substance	Single type of matter (elements, compounds)
Goods	Substances/mixtures with **economic values assigned by markets**; can include immaterial goods (energy, services, information)
Process	**Transport, transformation or storage** of materials (natural or man-made); e.g. raw material to vehicle, construction of highway
System	Set of material flows, stocks, processes within a **defined boundary** (city boundary, region, factory, household)
Activity	Set of systems needed to fulfil a **basic human need** (nourish, reside, transport etc.)
Stocks	Material reservoirs within the analyzed system
Flows	**Mass per time**
Fluxes	**Mass per time and cross section** (per unit area)
Imports/exports	Flows/fluxes **across system boundaries**
Inputs/outputs	Flows/fluxes **across process boundaries**

s11 diagram: system with extraction & production, delivery & shipment, consumption & use (stock), lithosphere (stock), landfill (stock); wastes, fuels, off-gas, emissions. s13: commercial life cycle – resource extraction → processing → manufacture → use → final disposal; Reduce, Recycle, Re-manufacture, Re-use loops; air/water/land outputs (green boxes).

Procedure of MFA s14–16

1. **Identification of key (material flow related) issues**
2. **System analysis** (selection of relevant matter, processes, indicator substances (elements), and system boundaries)
3. **Quantification of mass flows** of matter and indicator substances
4. **Identification of weak points** in the system
5. **Development and evaluation of scenarios**, representation, interpretation of results

[Order read from the colored chain diagram.] Flow chart (s15): problem definition → system definition (selection of substances, system boundaries, processes, goods) → determination of flows & stocks (mass flows; balancing of goods; concentrations; balancing of substances) → illustration & interpretation; feedback loops: refine mass flows, refine concentrations, redetermine goods, adjust system, redefine problem.

Classification of MFA s17–18

Based on	Parameter → example	Applications
Material considered	Unit flow of material – biomass, excavation; **Substance** – lead, iron, CFC, CO_2; **Product** – car, batteries	Development of environmental policy for hazardous substances; evaluation of product's environmental impact
System boundary	**Region** – city, municipality; **Sector** – transportation, construction; **Firm/industry** – car manufacturing plant	Firm environmental performance data; derivation of sustainability indicators; material flow accounts for official statistics

Applications s19–21

- **Environmental management:** environmental impact statements; remediation of hazardous waste sites; design of air pollution control strategies; nutrient management in watersheds; planning of soil-monitoring systems. (Ex.: clear picture of pollutant output from thermal plant/region → mitigation strategies.)
- **Resource & waste management:** analysis, planning, allocation, exploitation, upgrading of resources (ex.: limiting natural-resource exploitation, EV battery upgradation); modeling elemental compositions of wastes; evaluating material management performance in recycling/treatment facilities (output below threshold concentration of hazardous material).
- **Industrial ecology:** controlling pathways for materials use & industrial processes; creating **loop-closing** industrial practices; **dematerializing** industrial output; systematizing patterns of energy use; balancing industrial input/output to natural ecosystem capacity.

Advantages & limitations s22–24

- **Advantages:** critical view of resource management status in a city; evaluates environmental soundness of alternatives; decision tool to choose sustainable technology; ideal technical basis for planning & decision making especially in **developing and emerging countries with limited technical/financial resources**; suitable for **early detection of environmental problems** in developing countries.
- **Limitations:** needs a lot of data (only limited reliable data for developing countries); must deal with uncertainties due to dynamic nature of processes.

Numerical example s25–27

10 km stretch of 2-way 2-lane road. Absorbing capacity: vegetation **0.8** t CO_2/10000 m^2/h; road carriageway **0.2**; concrete surface **0.2**. Diagram labels: 10 km length; dimensions 180 m, 90 m, 70 m (which label refers to which width is not clear from the diagram). Traffic table (emission t CO_2/km; first lane/h; second lane/h): 2-wheelers 0.4 (10; 20); cars 1 (30; 15); small trucks 2 (15; 20); trucks 3 (5; 5); bicycle 0.1 (5; 5); e-cars 0.6 (1; 3).

- **Stock** = absorption by surface = [0.2×(2×90) + 0.2×70 + 0.8×(2×10)] × 24 = **1584 ton**.
- **Input** = CO_2 emission from vehicular movement = (emission per km × 10 km × vehicles/hr × 24) = **1640 ton CO_2/day**.
- **Output = Input − Stock = 1640 − 1584 = 56 ton CO_2/day**.

> Flag: the slide diagram dimensions (180 m, 90 m, 70 m, 10 m vegetation) are only partly legible as labels; the arithmetic above is copied from slide 27.

MFA vs EIA vs LCA s28–29

Method	Slide statements
LCA	Strives for **completeness** – as many substances as possible related to product; can be done for any specific product (e.g. aircraft, material used in construction)

MFA	Strives for **transparency and manageability** – limited number of substances; can establish an **inventory for an LCA** (e.g. fuel, EV battery input/output); assesses inflow & outflow of material for project area (system boundary); **doesn't analyse impacts**
EIA	Impact assessment of a **project**; generally considers impacts on **local environment after implementation**

Airport example: MFA studies input and outgoing materials; EIA assesses impacts of construction & operation activities.

s30 **Conclusion:** MFA assesses flow of materials within system; applications: environmental management, resource management, industrial ecology; compares alternatives → most resource-conservative option.

Important Distinctions to Remember (L36)

- **Flow** = mass/time; **flux** = mass/time/area. **Imports/exports** = across *system* boundaries; **inputs/outputs** = across *process* boundaries.
- **Substance** (single type of matter) vs **goods** (economic value) vs **material** (substances and goods). **System** vs **activity** vs **stock**.
- **MFA** = transparency/manageability, no impact analysis; **LCA** = completeness; **EIA** = project impacts on local environment.
- Stock = Input − Output.

L37 — MFA Case Study: Vienna (Austria) Urban Transport, Projections to 2050

Study area, background, need s3–5

- Vienna = capital and **most populous city of Austria (≈ 2 million)**; robust public transport.
- **Background:** in cities transport contributes significantly to global resource & energy consumption; **10–20% of Europe's urban population** exposed to excessive NO_2 (highest along busy roads); decarbonizing transport is an important means of mitigating climate change; studies focused almost exclusively on **energy and CO_2**, neglecting consumption of **material resources** needed for low-carbon transformation.
- **Need:** target to reduce Vienna's **per capita CO_2-eq emissions in transport by 50% by 2030 and by 100% by 2050**; propulsion technologies (EV) and shift to environment-friendly modes (walking, public transport); modal shift vs technological development are often **counter-poles** in decision-making → optimum scenario needed.

Objectives, research questions, system s6–11

- **Objectives:** effects of transforming urban transport on future **material stock and flows**; MFA of infrastructure and vehicles until 2050 under different scenarios (**characterized mainly by different modal splits**); time span **2016 to 2050**.
- **Research questions:** how will infrastructure & vehicle fleet develop to 2050; how do developments affect material stocks & flows; are there saving potentials for raw materials and how can they be realized.
- **System boundary:** transport system within administrative border of Vienna; infrastructure and vehicles; modes: **MIT** (motorized individual transport), **NMIT** (non-motorized individual transport; also written NMT), **PT** (public transport).
- **Materials** (in metric tons t, Mt): asphalt & bitumen, aluminum, batteries, brickwork, concrete, copper, glass, gravel & sand, iron & steel, rubber, plastics, wood (12 groups).
- **Flows & stocks:** construction & demolition of transport infrastructure; **maintenance** of infrastructure also considered; annual input/output flows computed from changes in stocks (increase/decrease); maintenance flows based on **renewal rates and useful lives**. **Not included:** material flows of **vehicle maintenance** and **network operation**.

Scenarios s12–15

Scenario	2016–30	2030–50
A – Business as usual (BAU)	Modal split **MIT 25%, PT 38%, NMT 37%**; already planned network extensions & vehicle procurement	Projection of prevailing system; EV technology of fleet stays at 2016 baseline
B – EV fleet	Same as A except **EV included in car fleet**	Car share **25%**; fossil cars replaced by electric/hydrogen; PT share **45%** (from 38%); NMT **30%** (from 37%)
C – Public Transport	Same as A except EV included in car fleet	PT **55%** (from 38%); MIT **<10%**; fleet by EV/hydrogen; NMT constant **35%**
D – Active Mobility	Same as A except EV included in car fleet	NMT distances increase (**>45%** from 37%); PT **45%** (from 38%); MIT **<10%**; fleet EV/hydrogen

Projections / results s16–27

- **Traffic modes:** highest personal-vehicle trips in **scenario B** (cars 6,240 million vehicle-km/year in 2050 vs 5,470 in A; **1,680** in C and D); least in C & D; most PT in C. 2020→2030: no major improvement in NMT/PT share.
- **Vehicle share:** drastic jump in EVs 2020→2030 & 2050; EVs also for commercial goods transport; **95% of motorbikes battery-based by 2050**. [2050 cars (scenario row): petrol 3%, diesel 3%, battery-electric 82%, hydrogen 10%; lorry <3.5 t BEV 95%, >3.5 t 50% – read from table.]
- **Active mobility:** share of bicycles reduced in B, C, D vs A and 2030 (replaced by **pedelecs = e-bicycles**); steady growth in 2050 scenarios except A.

- **Public transport:** grows drastically in 2050 vs 2020 in best case; **tram share of PT up to 47% in scenario D from 33% in 2020**.
- **Private motorized vehicle:** conventional vehicle use lower in 2030 than 2020; trips significantly reduced in **C & D** in 2050; people start walking/cycling for short distances with better PT & walking/cycle infrastructure.
- **Car fleet (B vs C):** petrol & diesel fleet **obsolete by 2050**; **70% fewer vehicles in C compared to B**; hybrid cars very few in both; in B, hydrogen vehicles increase rapidly after 2030.
- **Traffic area** = sum of **road, parking and walking/pedestrian** areas; road & parking area reduced in C & D vs A & B; traffic area reduced with PT/active mobility.
- **Rail network length** highest for **C & D**; same value for C and D affirms PT and NMT are **supplementary** to each other.
- **Transport construction material:** travel demand rises with time → material inflow rises; **scenario D least material usage**; **scenario B higher** due to equal traffic area but increased material for vehicle manufacturing. Per-capita inflow: significant MIT-material reduction in C & D vs A & B.
- **Cumulative infrastructure in/outputs:** flows similar for A & B, also for C & D; **outputs of C & D higher than A & B**; major shares: **sand/gravel, concrete, asphalt, steel**. **Vehicles:** major components **steel, aluminum, plastics**; output for C & D higher than input due to dismantling of existing road infrastructure [as stated].

Conclusions s28–31

- Choice of transport modes & motorization rate strongly affect material stock & annual resource consumption; **primary raw-material consumption reducible by shifting from private motorized vehicles to PT and active mobility**.
- If BAU retained, resource consumption & waste keep rising although GHG emissions might be mitigated (fewer fossil-fuel vehicles).
- **Greatest savings potential:** reduction in **private vehicle fleet and road infrastructure** and associated lower maintenance.
- To avoid negative feedback loops, **obsolete (road) infrastructure should be dismantled and converted** for other purposes; resource aspect should be considered in transport projects.
- MFA can evaluate mode-share scenarios; also use MFA besides direct impact evaluation; promoting PT and NMT helps resource conservation.

Important Distinctions to Remember (L37)

- Scenario keys: **A** = BAU; **B** = EV fleet (car share 25%, highest car trips & higher material); **C** = PT (PT 55%, MIT <10%, 70% fewer cars than B); **D** = Active mobility (NMT >45%, least material).
- MIT vs NMIT/NMT vs PT. Pedelec = e-bicycle.
- Maintenance of infrastructure **included**; vehicle maintenance & network operation **not included**.

L38 — Concept of Circular Economy

Economy & linear model s3–6

- **Economy:** concerned with production, distribution, consumption of goods & services; studies how individuals, businesses, governments, nations allocate resources; considered at national level but also local, regional, global; all economies interlinked ("global village"); general parameter = **GDP**; components: money supply, commercial activities/services, industry.
- **Linear model: take–make–consume–throw away**; based on "higher the consumption – bigger the economy".
- **Drawbacks of linear economy: 2–5% of global GDP lost due to congestion**; roads capture more urban land; worsening air quality; **one third of food produced globally (worth USD 1 trillion) wasted each year** while **more than 10% of world population goes hungry**. (Infographic s5: **10–15%** of building material wasted during construction; **60%** of European offices not used even in working hours; **20–40%** of energy in existing buildings can be profitably conserved; **54%** of demolition materials landfilled while some countries only landfill **6%**.)
- s7 Conceptual solution: minimize consumption & wastage; optimum utilization of resources; regenerate & reuse products/services; return resources to environment → model = **Circular Economy (CE)**.

Circular economy: definition, principles, features s8–12

- **Definition:** a **systemic approach** to economic development designed to benefit businesses, society and environment; a production & consumption model of **reusing, repairing, refurbishing, recycling** existing materials/products to keep materials within the economy wherever possible.
- **Three principles of CE: Design out waste and pollution**; **Keep products and materials in use**; **Regenerate natural systems**.
- **Features:** economy that is **restorative and regenerative by design**; economic activity builds & rebuilds overall system health; **waste becomes a resource** (minimizing waste); works at all scales (big/small business, organizations, individuals, global/local).
- **Linear vs circular: Take–make–dispose** vs **Make–use–return** or **make–consume–enrich**. [Diagram: linear – technical & biological materials mixed up, energy from finite sources; circular – biological & technical materials, energy from renewable sources.] Conceptual diagram stages: raw materials, design, production & remanufacturing, distribution, consumption/use/reuse/repair, collection, recycling, residual waste.

Ideas behind CE s13–23 (8 ideas in the concept diagram: performance economy, regenerative design, industrial ecology, ecological economics, cleaner production, cradle to cradle, blue economy, biomimicry)

Idea	Key facts
Performance economy (functional service economy)	Selling **service vs product**; based on product-life extension, long-life goods, reconditioning activities, waste prevention; products can be rented rather than owned
Cradle to Cradle	All material in industrial/commercial processes considered **nutrients**; design for effectiveness (positive impact; reduce negative impacts via efficiency); nature's **"biological metabolism"** as model for **"technical metabolism"**; components designed for continuous recovery & reutilization. **Based on 7 points:** eliminate the concept of waste; power with renewable energy; respect human & natural systems; manage water use to maximize quality; promote healthy ecosystems; respect positive local impacts; guide operations & stakeholder relationships using social responsibility
Biomimicry	**Janine Benyus** (book *Biomimicry: Innovation Inspired by Nature*): new discipline that studies nature's best ideas and imitates designs & processes to solve human problems; ex.: studying a **leaf to invent a better solar cell**; high-speed rail aerodynamic design. Nature as **Model** (emulate forms, processes, systems), **Measure** (ecological standard to judge sustainability of innovations), **Mentor** (value nature for what we can learn, not extract)

Industrial ecology	Study of material & energy flows through industrial systems; **closed-loop processes in which waste serves as input** (no undesirable by-product); systemic view, designing per local ecological constraints with global impact in view; industrial systems perform as close to **living systems** as possible; applicable to services sector
Natural capital	World's stocks of natural assets incl. **soil, air, water, all living things**; book *Natural Capitalism: Creating the Next Industrial Revolution*. **Principles (4):** radically increase productivity of natural resources; adopt nature's designs (outputs returned as nutrient or input for another process); move to **"service-and-flow" business model** (continuous flow of services vs sale of goods); reinvest in natural capital
Blue economy & regenerative design	Blue economy: **21 founding principles**, ocean-driven economy; solutions determined by local environment & physical/ecological characteristics; **gravity as primary source of energy**. **Regenerative design:** formulated for the **agriculture sector**; secondary products/waste used in other processes or as nutrients for natural systems
User vs consumer & sharing economy	Questions need to own products; **access to the service** a product provides matters, not the product; ex.: shared mobility can replace owning a car

Cascades, butterfly diagram, cycles s24–30

- **Cascades:** putting used material & components to different re-uses; **repetitive cycle until product returns to nature as nutrients**. Ex.: used paper recycled as paper/packing material/construction filling, then returned to nature (decomposable); cascades of wood.
- **Butterfly diagram** (Ellen MacArthur Foundation) = CE model. **Biological cycle** = **green, left side**: resource extraction from nature → cascade → biodegradation → nutrients for natural system. **Technological cycle** = **blue, right side**: technological materials that **cannot re-enter the environment** (metals, plastics, synthetic chemicals) must **continuously recycle** through the system so value is captured & recaptured.

Technological-cycle loop s30	Meaning
Maintain–prolong & share	Keep product & material in use by prolonging life span; design with durability & proper maintenance/repair; **shared among users** → reduces consumption
Refurbish & remanufacture	Restoring product value. **Refurbish** does **not disassemble** the product; cosmetic make-over. **Remanufacture:** whole product **disassembled into basic units**; necessary parts replaced before redistribution
Recycle	Reducing product all the way back to **basic material**; most or at least few parts remade into new products; **low value compared to reuse/refurbish** due to extra labor & energy, inevitable material loss, cost

Benefits s31–33

- **Economic:** job creation (labor-intensive recycling; higher-skilled manufacturing jobs; new service-based economy activities); innovation (recycling, remanufacturing, resource-savvy production); material cost saving; economic growth (increased revenues & more productive utilization of inputs).
- **Environmental:** contributes to global climate targets by designing out waste & pollution, keeping products/materials in use, regenerating rather than degrading natural systems.
- **For industries:** creation of new profit opportunities; reduced costs due to lower virgin-material requirements; stronger customer relationships via responsible/environment-friendly practices.

Linear → circular; Indian practices s34–38

- **System thinking:** business models, design, legislation, planning, practices, manufacturing all have undesirable qualities from circular view; **cannot change just one element** → need **holistic approach** rather than micro interventions.
- **Ancient practices in India:** domestic animals part of family; organic fertilizers (dung & organic waste) before chemical fertilizers; sacredness of trees; rivers, forests, earth treated like living being (now "system metabolism"); affection for ancestral property protects wells, resources, land.
- **Vedic Darshan:** "**Mata Bhumi putroham prithivyah**" = Earth is my mother, I am her son; Vedas hymns pray to Indra not to separate trees from forests and sons from fathers (**Rig-veda 8/1/13**); **Vṛkṣāyurveda:**

planting a tree equally beneficial as having **ten offspring**.

- **Buddhism:** environment not just to serve humans; humans only a part, no supremacy; **eco-centric** philosophy; per **Kutadanta**, ruler's responsibility to protect flora & fauna.
- **Tagore:** "highest education... makes our life in harmony with all existence". **Mahatma Gandhi: Gram Swarajya** – self-reliance & environment-friendly development of rural economy.

Conclusions s39

- Linear model → unsustainable; CE can replace it; CE based on older time-tested concepts; ensures environmental, economic, industrial benefits; holistic approach needed.

Important Distinctions to Remember (L38)

- **Linear** (take-make-consume-throw away) vs **circular** (make-use-return / make-consume-enrich).
- **Biological cycle** (green, left, nutrients/biodegradation) vs **technological cycle** (blue, right, metals/plastics/synthetic chemicals recycle continuously).
- **Refurbish** (no disassembly, cosmetic) vs **remanufacture** (full disassembly, parts replaced) vs **recycle** (back to basic material, low value).
- **Cradle to cradle** (waste = nutrients, 7 bases) vs **biomimicry** (model/measure/mentor) vs **industrial ecology** (closed-loop, waste = input) vs **blue economy** (21 principles, gravity) vs **regenerative design** (agriculture).

L39 — Circular Economy in Transport Sector

Need s3–4

- With growth, mobility systems face increased pressure → **negative externalities** without change of approach; transport is a key contributor to **emissions and resource consumption**; CE supports economic, environmental, social prosperity of city.
- Mobility = key urban priority; central to how a city operates; affects quality of life, local environment, resource consumption; effective systems accommodate **all modes**; CE optimizes transport by **integrating mobility planning with spatial planning** while improving products & services.
- **Four application areas:** vehicle design/manufacturing; transport infrastructure; mobility planning; transport-related policies.

CE in vehicle manufacturing s5–8

- Enable **repair, reuse, remanufacturing, recycling** of components & materials; design vehicles to be easily repaired in service and materials recovered at end of use; energy used to create parts is not wasted if parts are reused.
- **Strategies:** design for **adaptable and shared use**; maximum use potential & ability to remanufacture; **modularity** (parts swapped, space reconfigured) for longer use; **flexible interiors** in freight vehicles (simultaneous delivery & collection of goods).
- **Renault example:** materials (copper, steel, textiles, plastics) from **end-of-life vehicles (ELV)** put into new vehicles with same performance as virgin materials; **36%** of total mass of a newly produced Renault vehicle in Europe from recycled materials; **85% of an ELV is recyclable**. [Diagram "vehicle of tomorrow": renewable energy source, electric & silent propulsion, autonomous, long battery life, second-life battery options, designed for disassembly, re-manufactured locally, durable/upgradeable/easily repairable, 3D printed parts, connected, shared.]

CE in transport infrastructure s9

- Design infrastructure for adaptable use; construction materials with **minimum negative impacts and reusable** after life; **dynamic design** – ex.: dynamic street-use restrictions promoting specific modes at certain times (road space for bikes indicated by **LED lights in road**, freight-lane restrictions); simple low-cost interventions such as **moveable street furniture**.

CE in mobility planning s10–19

- **Compact city development** for effective mobility; infrastructure promoting walking, bicycling; **TOD** is also part of CE in mobility planning.
- **London Walking Plan:** by **2030 congestion projected to cost London GBP 9.3 billion a year**; walking action plan aims at **1 million additional walking trips a day**; invest **GBP 2.2 billion** (redesign streets, signposting & maps, more pedestrian crossings, better public transport); walkable London more land-use efficient, saves up to **GBP 1.6 billion** public healthcare costs, increases retail sales, reduces emissions, increases social cohesion.
- **Urban freight strategies:** optimized freight movement; coordination between delivery/logistic companies → minimum resources, lower emissions per kg. **Paris "logistics hotels"** (inner-city green freight): logistics facilities back in city in high-density areas; reduce heavy vehicle use, emissions, noise; parcels pooled from suburban logistics centres via **freight train or limited larger delivery shuttle-trucks**; deliveries in smaller lower-emission vehicles e.g. **electric tricycles carrying up to 180 kg**; space rented by city at favorable rate but requires low-emission modes.
- **Resource flows (capital & material):** plan mobility to minimize capital and resource use; a public transport system carries more people with less investment & fewer emissions.
- **China:** CE opportunities could reduce total cost of access to mobility by **CNY 12.6 trillion (USD 2 trillion) in 2030, an improvement of 36%** over current path; decrease in non-renewable resource (incl. fossil fuels) consumption by **49% in 2030 and 71% in 2040**.
- **São Paulo (reversing car-centric mobility): 2014** urban masterplan expands public & active transport; residents living near public transport from **25% in 2015 to 70% in 2025**; **30%** of urban development funds allocated; supports affordable housing and economic opportunities.

CE in policy & regulations s20–21

- Policies, regulations, incentives promote CE; vehicle recycling/scrappage incentives, high manufacturing standards.
- **Vehicle Scrappage Policy of India:** introduced in **Union Budget 2021-22**; formal regulations announced **August 2021**; ensures scientific scrapping; promotes material reuse; scrap value ≈ **4–6% of ex-showroom price** of new vehicle; promotes rapid shift **BS-IV → BS-VI**; incentive such as **no registration fee** on purchase of new vehicle; keeps polluting old vehicles out.

Case study: Kera, Espoo, Finland s22–33 (Paper: "Creating a Circular City")

- Partial case study; transport-related results used. **Kera** = new planned housing area in **Espoo, Finland**; typical example of **former industrial area rezoned into housing** with circular city targets; strategic target: international example of CE at city level. Study approach: alternatives → business models → regulations (realistic solutions for urban case district; examples of potential implementations; opportunities from regulatory perspective).
- **General data:** Kera area 79 ha (Espoo 52,800 ha); inhabitants 13,300 (Espoo 289,731); jobs 10,000 (120,676); residential floor area 532,000 m²; annual heating demand 43 GWh, cooling 4 GWh, electricity 30 GWh; transport: 115,693 (person) trips; average trip length 12.60 km.
- **Objectives (9):** carbon neutrality by **2030**; achieving UN SDGs; emission-free public transport by **2030**; enabling **MaaS (Mobility as a Service)**; deployment of electric buses; completely renewable energy production by 2030; emission-free district heating by 2030; effective utilization of waste as material & energy; promoting utilization of surplus food.

Scenario (2035)	Description
1	**Conservative**, relying on combustion-engine technologies; **deduced from 2035 figures for the whole of Finland**
2	**Accelerated adoption of carbon-free technologies**; penetration faster in largest cities (e.g. Espoo) because distances shorter and alternative refueling/charging networks develop favorably
3	**Ambitious future vision (CE concept adopted)**: public transport carbon-free by 2030; inhabitants early adopters → **all hybrid or electric passenger car base**

Mode / technology (s29)	Scenario 1	Scenario 2	Scenario 3
Public transport – Train (70%): electric	100%	100%	100%
Bus (30%): diesel / electric / gas	72% / 15% / 13%	40% / 40% / 20%	0% / 100% / 0%
Passenger car: gasoline / diesel / electric / plug-in hybrid / gas	40% / 18% / 20% / 14% / 8%	5% / 1% / 70% / 10% / 14%	0% / 0% / 78% / 15% / 7%

s30 Energy (kWh/pkm) & CO_2eq (g/pkm): cars – gasoline 0.37/93.61; diesel 0.35/82.93; gas 0.32/41.25; plug-in hybrid 0.17/23.40; electric 0.10/0.00; buses – diesel 0.17/41.35; gas 0.19/35.37; electric 0.06/0.00; train electric 0.08/0.00. s31–32 Annual energy consumption 2035 (GWh/yr): sums of table rows ≈ 100.8 (S1), 62.3 (S2), 51.0 (S3); annual carbon emissions (tCO_2eq/yr): sums of table rows ≈ 18,778 (S1), 5,038 (S2), 1,872 (S3) [computed by adding the slide's table rows].

- **Conclusions (s33):** in best case (**scenario 3**) transport energy consumption **could be halved** and emissions **decreased to 10% of the base level (scenario 1)**, mainly via **electrification**; best scenario depends on availability/cost of cleaner alternatives and on measures/incentives; significant share of gas-vehicle fuel could be produced locally from waste as energy resource.

Final conclusions s34

- CE can be adopted in **mobility planning, vehicle manufacturing, infrastructure development**; approaches: **TOD, EVs, walking & bicycling-friendly infrastructure**; using waste as energy for vehicles and recycling vehicles.

Important Distinctions to Remember (L39)

- CE in transport has **4 areas**: vehicle design, infrastructure, mobility planning, policies. Mobility planning sub-themes: compact city/TOD, urban freight, resource flows.
- Kera scenarios: **1 conservative** (Finland 2035 figures) · **2 accelerated carbon-free tech** · **3 ambitious CE** (all hybrid/electric cars; PT carbon-free by 2030).

- Recycling numbers: Renault **36%** recycled mass, **85%** ELV recyclable. Scrappage value **4–6%**.

Flag: s31–32 charts were read from rendered images; totals are my sums of the slide's table rows.

L40 — Modelling of Transport Emissions-I

Vehicular emissions & inventory s3–8

- **Three types of vehicular emissions: Exhaust**, **Evaporative**, **Non-exhaust** (tyre and brake).
- **Exhaust:** CO, HC, NOx, CO_2, PM, benzene, 1-3 butadiene, formaldehyde, PAH (polycyclic aromatic hydrocarbons), acetaldehyde, total aldehyde, GHG emissions. **Evaporative:** VOCs/hydrocarbons. **Non-exhaust:** tyre & brake emissions – **PM2.5 and PM10**, heavy metals.
- **Emission inventory** = important tool to assess vehicular emissions based on **emission factors, average speed, fuel consumption and amount of traffic** on a defined road type; developed manually or by computer-based models. **Models developed on:** vehicle type; fuel category & engine technology; vehicle kilometers travelled (VKT); emission standards; various correction factors.
- Decision Support System (DSS) framework: model-based emission inventory development → data analysis/action to implement policy → traffic development & policy implementation → evaluation of efficiency of policies. Emission model = summation of product of emissions from all vehicular sources, vehicular activities and climatic & topographic variables.
- **Model inputs:** vehicle population, type, age, model, technology, fuel type, VKT, temperature, humidity, altitude. **Outputs:** emission inventory by pollutants; vehicle-type contribution; efficiency of current policy; input data for dispersion models; vehicular emission trends; fuel consumption & contribution.

Classification of transportation emission models s9–10

Static (Top-Down, macro-scale) = aggregated emission factor models, average speed models. **Dynamic (Bottom-Up, micro-scale)** = traffic situation models, instantaneous models. Top-down works on macro scale only; bottom-up uses detailed micro-level data.

	Dynamic		Static	
Model	**Traffic situation**	**Instantaneous**	**Average speed**	**Aggregate emission factor**
Base of modeling emission factors	Speed & traffic factors	Engine power	Average speed of trip	VKT, transport productivity or fuel consumption
Scale	Micro	Micro	Macro	Macro
Usage	Street-level emission	Street-level	National or strategic emission estimation	Emissions of unregulated pollutants
Limitations	Needs detailed statistics about speed & traffic situations	Needs detailed and expensive data	Neglects cycles & driving behaviours of same average speeds	Cannot be used for situations not explicitly defined

Vehicular emission models s11–38

Examples (s11): MOBILE series (USEPA); MOVES (USEPA); EMFAC (CARB); COPERT (EEA); QGEPA model (Australia); IVE model (funded by USEPA). s12 diagram: road vehicles → movement of goods/passengers; fuel in; evaporative & exhaust emissions; road-vehicle tyre & brake wear; road wear.

Model	Key facts
MOBILE (USEPA) s13–15	Highway-vehicle emissions in the **U.S. except California**; **MOBILE6.2** last version (**2004**), later updated to/superseded by **MOVES**; calculates **HC, NOx, CO** from cars, light & heavy duty vehicles; uses ambient temperature, humidity, altitude; basic emission rates for average speed of **31.5 km/h** for light-duty gasoline vehicles by model year & technology; computes hot running emissions per unit distance for **5 mile/hr speed increments**: **ex = SCF (BER + EO)** – SCF speed correction factor, BER basic emission rate, EO emission offset (correction for "off-cycle" emissions – additional emissions due to high power operation not in BER)

MOVES (MOtor Vehicle Emission Simulator) s16–18	Currently **state-of-the-science** emission system of USEPA; mobile sources across U.S. except California; air pollutants, GHGs, air toxics; **MOVES3 latest** (after MOVES2010, MOVES2014); emission types: engine running, starting, **hoteling (extended idling)**, evaporative, brake & tyre; inputs: default national averages – vehicle counts, **VMT (Vehicle Miles Travelled)**, temperature, fuel, location etc. Running-emission calculation inputs: vehicle miles travelled, average speed, driving cycle, source hours operating, operating mode distribution, emission rates (g/hr), adjustments (fuel parameters, temperature, humidity, A/C) → adjusted emissions; summed by model year, regulatory class, fuel type
EMFAC (California Air Resources Board, CARB) s19–23	Emission inventories of **on-road mobile sources in California**; input from **CALIMFAC** (California Motor Vehicle Emissions Factor Model: emission rates and weight for vehicle activity by model year); vehicle characteristics (VKT, starts, number of vehicles) provided by **BURDEN** model; **latest EMFAC2021**. Equation: **ex = BER × SCF = BER × EXP(b1(v–27.4) + b2(v–27.4)²)**; BER = mean emission factor of individual vehicle (total g in lab test, unified cycle length in mile); SCF developed from **12 driving cycles**; b1, b2 model parameters; v in mile/h. EMFAC2021 vs MOVES3 evaporative emissions (passenger cars, Los Angeles county): similar trends & magnitudes; EMFAC2021 uses California-specific inputs. NOx start emissions (heavy-duty diesel trucks): EMFAC2021 higher than EMFAC2017 especially for **soak times < 400 minutes**. (Soak time = duration engine is off [slide text cut off].)
COPERT (European Environment Agency, EEA) s24–29	Developed by EEA; **CORINAIR** working group (first European initiative on road-traffic emissions) transformed to COPERT in **1989**; **COPERT 5.4.52** latest (May 2021); computes hot running, evaporative, non-exhaust (brakes, tyres); **105 vehicle categories in 5 classes** (cars, light & heavy duty vehicles, urban buses & coaches, two wheelers); pollutants: CO, NOx, VOC, CH_4, CO_2, N_2O, NH_3, SOx, diesel PM, PAHs, POPs, dioxins, furans, heavy metals (lead, cadmium, copper, chromium, nickel, selenium, zinc). **E_TOTAL = E_HOT + E_COLD + E_EVAP**; also E_TOTAL = E_URBAN + E_RURAL + E_HIGHWAY (driving conditions); E_EVAP only relevant for **NMVOC from gasoline vehicles**; E_COLD = transient thermal engine operation (cold start). **New in 5.4:** plug-in hybrid (PHEV) passenger cars; hybrid buses; revised Euro 5 & Euro 6 factors for passenger cars & LCVs; parameter updates/error fixes. PHEV in charge-sustaining (CS) mode: petrol PHEV → equation for petrol hybrid; diesel PHEV → equation for diesel conventional
QGEPA (Queensland Govt Environmental Protection Agency, Australia) s30–31	Travel-speed emission model; hot running emission factor database for **CO, HC, PM, NOx in g/km**; classes: light duty (petrol, diesel, LPG), heavy duty (diesel, CNG, LPG), motorcycles; separate factors for cold start and evaporative; also non-regulated pollutants (hydrocarbons, N_2O, NH_3, PAH); speed correction factors used for evaporative running loss emissions for cold start
IVE (International Vehicle Emissions) s32–38	Developed jointly by **UC Riverside CE-CERT, GSSR, ISSRC**; **funded by USEPA**; accounts for technologies & conditions in **developing countries** (driving patterns, **vehicle specific power (VSP)**, engine stress distributions); inputs: type & age of fleet, local conditions (temperature, pressure), fuel specifications; estimates criteria pollutants, GHGs, toxics: CO, VOC, NOx, PM2.5, PM10, CO_2, N_2O, CH_4, NH_3, benzene, lead, 1,3-butadiene, aldehydes; emissions from a single roadway or entire area for specific time periods. **Two basic equations: Eq. 1** Q_T = B(t) × K(base) × K(temp) × K(Hmd) × K(IM) × K(Alt) × K(Cntry) – adjusted emission rate = base rates × correction factors; **Eq. 2** Q_running = U[FTP] × D/Uc × Σ_t{f[t] × Q[t] × Σ_d{f[dt] × K[dt]}} – weights adjusted rate by travel fraction per technology & driving type; final step multiplies by ratio of **average velocity of LA-4 driving cycle** to average velocity of modeled cycle and by distance (running emissions only); result = overall fleet running emissions (g). U.S. **FTP-72** cycle also called **UDDS** or **LA-4** cycle. Variables: B[t] base rate; f[t] fraction of travel by technology; f[dt] fraction of each driving type; K(Tmp), K(Hmd), K(Alt), K(Fuel), K(IM) inspection/maintenance, K(Cntry), K[dt] driving correction factor; D distance/day (km)

Indian context s39–42

- **Pune example:** dominant vehicles two-wheelers followed by three-wheelers; **buses and light-duty four-wheelers major NOx source**.
- **Model complexity & errors:** reliability depends on **accuracy and robustness**; accuracy improved by more complex formulations, but robustness tends to be negated by increasing input errors.
- **Limitations for India:** MOBILE, MOVES, COPERT etc. developed in **developed countries (US, Japan, Australia)** with extensive datasets; differences in technology, infrastructure, driving cycle → may give

erroneous results; emission factors/basic emission rates are **US & Europe based**; IVE designed for developing countries like India but **complexity similar to US/European models** (extensive data, hard to compile); most models cannot give output for **more than a year**; need experimental data not available for India; most are **region specific**.

VAPI model (Vehicular Air Pollution Inventory, IIT Roorkee) s43–54

- Designed specifically for **developing countries like India**; incorporates **climatic correction factors**; fair agreement with Indian air-quality observations; **simple to use, public domain**.
- **Assumptions:** vehicles per 1000 persons grows in a **Sigmoidal (S-shaped)** relationship; saturation = **one car plus two wheelers per family**; seating capacities of auto-rickshaws & taxis **3 and 5** → saturation per 1000 persons **330 and 200**. Vehicle population projection based on **GDP and per capita income** using the **Gompertz model**; on-road population = projected past-year population subtracted from cumulative registered population; base emission factors depend on vehicle age and model year.
- **Pollutants:** running exhaust – CO, HC, NOx, CO_2, PM, benzene, 1-3 butadiene, formaldehyde, total PAH, acetaldehyde, total aldehyde; **running loss evaporative** emissions (VOC); **non-exhaust** (brakes, tyres) – PM10 & PM2.5.
- **Running exhaust:** $EF_{a,p,i} = EF_{p,i} \times C_T \times C_H \times C_A$ (temperature, humidity, altitude correction factors); $E_{t,p,i} = P_i \times EF_{a,p,i} \times V_i \times D_{t,i}$ (P = on-road population; V = per day distance km/day; D = annual travelling days). **Evaporative:** $EF_a = EF \times C_T \times C_A$ (temperature & altitude only); running losses from petrol vehicles are largest source of VOC evaporative emissions (others: diurnal, hot soak, resting losses); running losses = vapor generated in gasoline tanks during vehicle operation. **Non-exhaust PM:** generated mechanically by abrasion of tyre, brake, road pavement and resuspension of road dust.

Conclusion s55

- Models mainly developed by US, Europe, Australia; complex, predict emissions of all vehicle categories; but mostly **country specific** and based on extensive native datasets; **VAPI is a simple model designed for developing countries like India**.

Important Distinctions to Remember (L40)

- **Static = top-down = macro** (aggregate emission factor, average speed) vs **dynamic = bottom-up = micro** (traffic situation, instantaneous).
- **MOBILE** (superseded; US except California) → **MOVES** (current; US except California) vs **EMFAC** (California) vs **COPERT** (EEA, Europe) vs **QGEPA** (Queensland, Australia) vs **IVE** (developing countries; UC Riverside etc.; USEPA-funded) vs **VAPI** (IIT Roorkee, India).
- **MOBILE ex = SCF (BER + EO)** vs **EMFAC ex = BER × SCF** (exponential, 27.4, 12 driving cycles) vs **COPERT E_TOTAL = E_HOT + E_COLD + E_EVAP**.
- VAPI evaporative uses temperature + altitude only; exhaust uses temperature + humidity + altitude.

Flags: L40 s21, s26 equations and s47–52 VAPI equations 3–17 (images) partially legible; only equations stated in slide text/legible images are given. s26 slide text says "calculated by the EMFAC" for the COPERT equation [as written on slide]. s23 slide text ends mid-sentence. s12, s18, s34–35, s45, s48–52 are diagrams/equation images; contents summarized as legible.

Part 3 — Rapid-Revision Sheet

L31–L32: Planning, calming & TSM

- Sustainable transport = least damageable to environment + doesn't impair future generations' mobility. Planning preference: **accessibility > mobility**; early planning = induced demand; traffic problems rose since **1970s**.
- Planning process 5 stages: goals → options → actors/stakeholders (public participation) → predict outcomes & costs (critical) → choose action (political).
- Forecast flaws: A under-estimation (early), B over-estimation (late), C "return to normal" bias. Path dependency = locked in. Megalopolis examples: BostWash, Tokaido, Pearl River Delta, Shanghai–Nanjing.
- Calming: vertical deflection, horizontal shift, narrowing (→ speed); closures (→ cut-through). Hump ≈ 30 km/h; bump more aggressive; table = flat top; choker = mid-block curb extension; lateral shift/chicane ≤ 35 mph; road diet 4→3 lanes (TWLTL); Seattle Rainier Ave −95% speed.
- TSM = most productive, cost-effective use of existing services. T-turn / G-turn / Q-turn. One-way = least expensive. Van pool 7–15. Car pool = 2nd most common US work mode; HOV lane min 3 riders. Staggered hours: core time fixed; 1 SEK = 8.61 INR. Congestion pricing: London 2003 cordon, Singapore. London CPZ Mon–Fri 7–7, Sat 7–11. Delhi–Gurugram 14 km, trucks barred 8–10 AM & 5–10 PM. Exclusive bus lane: min 3 lanes, next to curb (Ahmedabad). Odd-even fails: second cheap car. Jakarta: 2 Sundays/month.

L33–L35: LCA

- LCA = entire life cycle; 4 objectives (product improvement, decision making, environment planning, marketing); 4 stages (goal & scope → LCI → impact assessment → interpretation).
- Boundaries: cradle-grave (linear) · cradle-cradle (closed-loop, recycling) · cradle-gate (production) · gate-grave (use+end-of-life) · gate-gate (production only) · well-to-wheel (fuels/vehicles) · EIOLCA (sector averages).
- Scope 8 items; functional unit vs reference flow; allocation: mass / heating value (fuels) / market value. LCIA steps (EPA, 7): categories, classification, characterization, normalization, grouping, weighting, evaluating & reporting. Impact categories: global warming, acidification, eutrophication, smog, ozone depletion.
- ISO: 165 bodies, 1947. 14040 principles · 14041 goal/scope+LCI · 14042 LCIA · 14043 interpretation · 14047 LCIA examples · 14048 data formatting · 14049 examples for 14041.
- Types: conceptual (life cycle thinking) · simplified (screening, simplifying, reliability) · detailed (Consoli 1993: planning, screening, data collection & treatment, evaluation, improvement assessment).
- Milan/Turin: LTZ; Milan 8.2 km², Turin 2.58 km²; baseline diesel vs ZEV; GHG −37% / −50%; PM +13% / +4%; NMVOC ≈ −50%; 50% PV → comparable; 100% solar → PM −7%; Turin all EURO 5, Milan EURO 4+5.
- Denmark: 100% renewable 2050; wind up to 50%; 2008 fossil 84% / renewable 16% / oil 39%; transport 34%, households 29%, industry 18%; scenarios 2008, 2030, 2050 CSV/RME/BtL; functional unit = meeting energy demand; horizon 100 yr; GW 68→38→20/31/17; fossil diesel preferred except global warming (FT-biodiesel).

L36–L37: MFA

- MFA = flows & stocks of materials in system defined in space & time (Brunner & Rechberger 2004); Input = Output + Stock; Lavoisier 1789; Greek philosophers >2000 yrs. Flow = mass/time; flux = mass/time/area. Procedure 5 steps. Classification: material (flow/substance/product) & system boundary (region/sector/firm). Lead example 200 − 150 = 50 g. Numerical: stock 1584, input 1640, output 56 t CO_2/day.
- MFA (transparency, no impact analysis; can form LCA inventory) vs LCA (completeness) vs EIA (local project impacts).
- Vienna: ~2 million; 2016–2050; materials in t; 12 material groups; MIT/NMIT/PT; BAU modal split 25/38/37; B car 25%, PT 45%, NMT 30%; C PT 55%, MIT <10%, NMT 35%; D NMT >45%, PT 45%;

95% motorbikes battery; tram 33%→47%; C = 70% fewer cars than B; D least material; vehicle maintenance & network operation excluded; 50% by 2030 & 100% by 2050 CO_2 target.

L38–L39: Circular economy

- Linear = take-make-consume-throw away. CE principles: design out waste/pollution; keep products & materials in use; regenerate natural systems. Ideas: performance economy, cradle to cradle (7), biomimicry (model, measure, mentor; Benyus), industrial ecology, natural capital (4 principles), blue economy (21 principles), regenerative design (agriculture), sharing economy, cascades. Butterfly: biological (green, left) / technological (blue, right). Refurbish (no disassembly) vs remanufacture (disassembled). Stats: 2–5% GDP lost to congestion; 1/3 food wasted (USD 1 trillion); >10% hungry.
- Transport CE: Renault 36% recycled, 85% ELV recyclable; London walking: £9.3 bn congestion by 2030, 1 million trips, £2.2 bn, £1.6 bn health; Paris logistics hotels (tricycles 180 kg); China CNY 12.6 trillion (36%), 49% (2030)/71% (2040); São Paulo 25%→70% (2015→2025), 30% funds; India scrappage 2021-22, 4–6% scrap value; Kera (Espoo): 3 scenarios, S3 energy halved, emissions 10% of S1.

L40: Emission models

- 3 emission types (exhaust, evaporative, non-exhaust). Static/top-down/macro vs dynamic/bottom-up/micro. MOBILE (MOBILE6.2, 2004; 31.5 km/h; 5 mph increments) → MOVES (MOVES3). EMFAC (CARB; CALIMFAC; BURDEN; EMFAC2021; 12 cycles; 27.4). COPERT (EEA; CORINAIR 1989; 5.4.52; 105 categories, 5 classes). QGEPA (g/km). IVE (developing countries; VSP; LA-4). VAPI (IIT Roorkee; Gompertz; sigmoidal; saturation 330 & 200 per 1000 for auto-rickshaw & taxi; seats 3 & 5).

Part 4 — Practice MCQs (30)

1. (L31) Which traffic-calming measure is intended to reduce cut-through traffic rather than speed?
A. Vertical deflections B. Horizontal shifts C. Closures D. Narrowing roadways

2. (L31) A curb extension placed at a mid-block location is called a:
A. Raised crosswalk B. Choker C. Plateau D. Offset speed table

3. (L31) According to the slides, the "Return to normal" bias is flaw number … in forecasting-based planning:
A. A (early-stage under-estimation) B. B (late-stage over-estimation) C. C D. None of these

4. (L32) A diversion of right-turning traffic to the left before the junction is a:
A. T-turn B. Q-turn C. U-turn D. G-turn

5. (L32) Exclusive bus lanes are possible only where the carriageway has a minimum of:
A. 2 lanes B. 3 lanes C. 4 lanes D. 5 lanes

6. (L32) Which statement about the Delhi–Gurugram Expressway truck restriction is correct?
A. 14 km stretch; no trucks 8–10 A.M and 5–10 P.M B. 40 km stretch; no trucks 8–10 A.M C. 14 km; no trucks 7 A.M–7 P.M D. 4 km; no trucks at night

7. (L33) The LCA system boundary that covers only the production phase is:
A. Cradle to gate B. Gate to grave C. Gate to gate D. Cradle to cradle

8. (L33) Allocation "by heating value" is often used for production processes of:
A. Fuels B. Cement C. Plastics D. Vehicles

9. (L33) Which is the correct order of the LCA stages?
A. LCI → Goal & scope → Impact assessment → Interpretation B. Goal & scope → LCI → Impact assessment → Interpretation C. Goal & scope → Impact assessment → LCI → Interpretation D. Interpretation → LCI → Goal & scope → Impact assessment

10. (L34) ISO 14043 provides:
A. Examples for applying ISO 14041 B. Guidance on interpretation of LCA results C. Guidance on conducting LCIA D. Information on formatting of data

11. (L34) In the Milan–Turin study, adopting the ZEV fleet with the current electricity mix resulted in particulate matter emissions that:
A. Fell 13% in Milan and 4% in Turin B. Rose 13% in Milan and 4% in Turin C. Rose 37% in Milan and 50% in Turin D. Did not change

12. (L34) The simplified LCA stages are:
A. Planning, screening, evaluation B. Screening, simplifying, reliability C. Goal, inventory, interpretation D. Screening, planning, improvement

13. (L35) In 2008, the share of fossil fuels in Denmark's primary energy supply was about:
A. 16% B. 39% C. 84% D. 100%

14. (L35) In the Denmark study, which fuel was favorable to biomass-based fuels for heavy transport because of large upstream land-use-change impacts?
A. Fossil diesel B. Bioethanol C. Biogas D. Syngas

15. (L35) The "BtL" 2050 scenario uses:
A. Biodiesel entirely from rapeseed B. Biomass-to-Liquid with Fischer–Tropsch technology C. A conservative fossil-fuel scenario D. Only bioethanol

16. (L36) In MFA, "flux" is:
A. Mass per time B. Mass per time and cross-section C. Flows across system boundaries D. Material reservoirs within the system

17. (L36) In the numerical example, the output is:
A. 1584 ton CO_2/day B. 1640 ton CO_2/day C. 56 ton CO_2/day D. 3224 ton CO_2/day

18. (L36) Which method strives for completeness and considers as many substances as possible?
A. MFA B. EIA C. LCA D. TSM

19. (L37) Which Vienna scenario has the least material usage?
A. A (BAU) B. B (EV fleet) C. C (Public Transport) D. D (Active Mobility)

20. (L37) In scenario C (2030–50), the PT share rises to:
A. 38% B. 45% C. 55% D. 25%

21. (L37) Which is NOT included in the Vienna study's material flows?
A. Construction of infrastructure B. Demolition of infrastructure C. Maintenance of infrastructure D. Vehicle maintenance

22. (L38) The technological cycle of the butterfly diagram is shown:
A. In green on the left B. In blue on the right C. In green on the right D. In blue on the left

23. (L38) Which process disassembles the whole product into basic units and replaces necessary parts before redistribution?
A. Refurbish B. Recycle C. Remanufacture D. Share

24. (L38) The three principles of circular economy are design out waste and pollution, keep products and materials in use, and:
A. Maximise consumption B. Regenerate natural systems C. Take-make-dispose D. Sell products rather than services

25. (L39) Currently, what percentage of the mass of a newly produced Renault vehicle in Europe is made from recycled materials?
A. 85% B. 36% C. 49% D. 71%

26. (L39) In the Kera study, Scenario 3 is described as:
A. Conservative, based on 2035 Finland figures B. Accelerated adoption of carbon-free technologies C. Ambitious future vision with CE adopted D. Baseline of 2008

27. (L39) India's vehicle scrappage value is stated to be around what share of ex-showroom price of a new vehicle?
A. 4–6% B. 10–15% C. 25% D. 36%

28. (L40) Which model was developed by the California Air Resources Board?
A. COPERT B. MOVES C. IVE D. EMFAC

29. (L40) Static (top-down) emission models include:
A. Instantaneous models B. Traffic situation models C. Average speed models D. None

30. (L40) The VAPI model assumes saturation levels per 1000 persons for auto-rickshaws and taxis of:
A. 200 and 330 B. 330 and 200 C. 3 and 5 D. 5 and 3

Answer key

Q	Ans	Source-based explanation	Slide
1	C	Closures reduce cut-through traffic; the other three reduce speed.	L31 s26
2	B	Curb extension at mid-block = choker.	L31 s31
3	C	Label C = "Return to normal" bias.	L31 s3
4	D	G-turn = diversion to the left before junction; Q-turn = beyond junction; T-turn = alternative intersection.	L32 s5
5	B	Minimum 3 lanes.	L32 s44
6	A	14 km; 8–10 A.M and 5–10 P.M.	L32 s35
7	C	Gate to gate = production phase only.	L33 s16
8	A	Heating-value allocation often used for fuels.	L33 s28
9	B	Goal & scope, LCI, impact assessment, interpretation.	L33 s21
10	B	14043 = interpretation of results.	L34 s5
11	B	PM +13% Milan, +4% Turin.	L34 s30
12	B	Screening, simplifying, reliability.	L34 s10
13	C	Fossil ≈ 84%; renewable 16%; oil 39%.	L35 s6
14	A	Large LUC impacts made fossil diesel favorable to RME- and FT-biodiesel.	L35 s32
15	B	BtL = Biomass-to-Liquid, Fischer–Tropsch.	L35 s14

16	B	Fluxes = mass per time and cross section.	L36 s12
17	C	Output = 1640 − 1584 = 56.	L36 s27
18	C	LCA strives for completeness; MFA for transparency & manageability.	L36 s28
19	D	Scenario D has least material usage.	L37 s24
20	C	Scenario C PT 55% (from 38%).	L37 s14
21	D	Vehicle maintenance & network operation not part of investigation.	L37 s10–11
22	B	Technological cycle: blue, right-hand side.	L38 s28
23	C	Remanufacture disassembles whole product; refurbish doesn't.	L38 s30
24	B	Regenerate natural systems.	L38 s9
25	B	36% (85% of an ELV is recyclable).	L39 s7
26	C	Scenario 3 = ambitious future vision (CE adopted).	L39 s28
27	A	≈ 4–6% of ex-showroom price.	L39 s21
28	D	EMFAC – CARB.	L40 s19
29	C	Static = aggregated emission factor + average speed models.	L40 s9
30	B	330 and 200 (seating 3 and 5).	L40 s44

Part 5 — Coverage Audit

Slide counts were checked with pdfinfo (total 424). Every slide's text was extracted and read; picture-only slides were rendered and viewed. Reference slides list citations only and are not reproduced. Slide s2 "Contents" slides are agenda lists already reflected in the notes' structure.

PPT filename	Total slides	Slide numbers covered in notes	Extraction issues
lec 31.pdf	64	1–64 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s3,7,15,19–21,29 image slides read and transcribed; s12, 18, 32–36, 38, 40, 42–43, 45–46, 48, 50, 52, 54–55, 57, 59–61 are photos/illustrations (labels noted where legible). No unreadable content.
lec 32.pdf	50	1–50 (all; s1 title, last-two reference/thank-you slides contain no testable content)	Images read: s7, 12, 16, 28 (values given); s5, 9, 15, 18, 37, 40, 45 diagrams summarized. Photo-only slides (s6, 21, 24–25, 41–43, 46) carry only captions already noted.
lec 33.pdf	40	1–40 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s20, 31, 35, 37 diagrams/tables read; s4, 9, 11–14, 25, 30 are illustrations (captions noted).
lec 34.pdf	34	1–34 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s22 vehicle-data table (small print) summarized by attributes only, not row-by-row; s26 table read; s27–28 charts summarized from legends. s13 title image only.
lec 35.pdf	37	1–37 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s15–20 Sankey energy-flow diagrams: individual flow numbers only partly transcribed (CSV/RME/BtL key values given); s27 small table: totals only. Charts s28–31 read.
lec 36.pdf	32	1–32 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s11, 13–16, 25–26 read. Dimension labels on s26 diagram (180/90/70 m) ambiguous – flagged.
Lec 37.pdf	33	1–33 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s16 and s17 small tables: only highlighted/stated values transcribed; s18–27 charts summarized from bullets and legends.
Lec 38.pdf	41	1–41 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s10, 12, 23, 26, 29 diagrams read; s25 'Cascades Wood' image summarized by title only; s30 text is embedded in a flowchart image, read.
Lec 39.pdf	36	1–36 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s24, 27, 29–32 read (table totals computed); s8, 19 diagrams summarized.
Lec 40.pdf	57	1–57 (all; s1 title, last-two reference/thank-you slides contain no testable content)	s21, 26, 47–52: equation images partly legible; VAPI equations 3–17 (s48–52) not transcribed in full – flagged for manual review. s10, 18, 34–38, 45 read.

Limitations disclosed: (1) Sankey diagrams (L35 s15–20) and equation images (L40 s47–52) are not fully transcribed. (2) Photos/illustrations that carry no text are described only by caption. (3) Where the slides contradict themselves (e.g. L34 Turin climate change −49% in table vs 50% in text) both are shown. (4) Values read off charts are marked [read from graph]. Items flagged above should be checked manually against the PDFs.